

*Heaven's Light is Our Guide*

## **Rajshahi University of Engineering & Technology**



Department of Computer Science & Engineering

24 Series

Course Code: CSE 2100

Course Title: Software Development Project I

**Title:** Software Project Proposal

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# CampusOS

## Problem Statement:

University students and staff often rely on multiple communication platforms such as Messenger, WhatsApp, Telegram, Google Classroom and the university website to receive academic and administrative updates. This fragmented communication system frequently leads to missed announcements, confusion, unnecessary distractions and wasted time

## Objective:

The objective of **CampusOS** is to develop a centralized desktop application that streamlines academic communication by providing students and university authorities with a single platform for notices, course materials, schedules, deadlines and important announcements

## Project Description:

**CampusOS** is a role-based campus management application designed for students and university authorities. After logging into the system, each user is automatically redirected to a personalized dashboard according to their role. Students can easily access notices, lecture materials and deadlines, while authorized personnel such as Class Representatives (CRs), Teachers, Advisors and Heads of Department (HOD) can publish announcements and manage academic communication. As a result, instead of switching between several different applications every day, users receive all important information through one organized dashboard

## Target Users:

- Students
- Class Representatives (CRs)
- Teachers
- Advisors
- Head of Department (HOD)
- (Future) Registrar Office, Library Staff, Examination Controller

## Features:

- Secure login with role-based authentication
- Separate dashboards for different user roles
- Notice creation, editing and deletion
- Pinned announcements
- Five categorized notice tabs (All, CR, Teachers, Advisors, HOD)
- File upload, preview and download (PDF, Images, Documents)
- Search notices using keywords
- Deadline reminder notifications

- User profile management
- Responsive and user-friendly interface
- SQLite database integration
- Dark mode

### **Technologies:**

- **Programming Language:** Java
- **GUI Framework:** JavaFX
- **UI Design:** FXML & Scene Builder
- **Database:** SQLite
- **Database Connectivity:** JDBC
- **IDE:** Visual Studio Code
- **Version Control:** Git & GitHub

### **Expected Outcome:**

**CampusOS** will provide a unified communication platform that improves information accessibility, reduces confusion, minimizes missed announcements and increases overall productivity for students and academic staff

### **Future Scope:**

- Mobile application
- Cloud synchronization
- AI-powered notice summarization
- Academic calendar integration
- Attendance management
- Assignment submission system
- Real-time notifications
- Chat system
- Read analytics and message tracking
- Multi-university deployment

### **Justification:**

**CampusOS** addresses a real communication problem experienced by university students and faculty members. It demonstrates practical software engineering concepts such as authentication, role-based access control, database management, graphical user interface development and modular software design, making it an ideal academic software project

# ExpiryWise

## Problem Statement:

Many households waste food because people forget expiry dates or lose track of the items stored in refrigerators, freezers and pantries. This not only causes unnecessary financial loss but also contributes significantly to food waste

## Objective:

The objective of **ExpiryWise** is to help users organize their food inventory, monitor expiry dates, receive timely reminders and reduce food waste through a simple and user-friendly application

## Project Description:

**ExpiryWise** is a food inventory management application that allows users to record food items together with their expiry dates and storage locations. The application automatically categorizes food according to its freshness status and sends reminder notifications before items expire. Users can search, filter, edit and organize their inventory through a simple dashboard while viewing useful statistics regarding food usage and waste

## Target Users:

- Students
- Families
- Individuals
- Small restaurants
- Anyone who wants to reduce food waste

## Features:

- User login system
- Add new food items
- Edit and delete food items
- Food categories
- Search and filter functionality
- Automatic expiry status (Safe, Use Soon, Expires Today, Expired)
- Dashboard showing food statistics
- Reminder notifications before expiry
- Calendar view
- Food analytics
- Shopping list
- SQLite database integration
- Dark mode

## **Technologies:**

- **Programming Language:** Java
- **GUI Framework:** JavaFX
- **UI Design:** FXML & Scene Builder
- **Database:** SQLite
- **Database Connectivity:** JDBC
- **IDE:** Visual Studio Code
- **Version Control:** Git & GitHub

## **Expected Outcome:**

**ExpiryWise** will help users reduce food waste, save money, improve food organization and receive timely reminders before products expire, creating a more efficient household food management system

## **Future Scope:**

- Barcode scanner
- AI recipe recommendations
- Cloud backup
- Firebase synchronization
- Family sharing
- Export to PDF/Excel
- Multi-language support
- Gamification and achievement system

## **Justification:**

**ExpiryWise** solves a practical real-world problem while incorporating important software engineering concepts such as database management, notifications, search, filtering, graphical dashboards and user authentication. The modular design also allows future expansion with AI and cloud-based features